UDAY PRAKASH LAKKARAJU

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EDUCATION

Stevens Institute of Technology New Jersey, USA

Master of Science in Information Systems | GPA: 3.6 ***Sep 2024 — May 2026***

Relevant Coursework: Business Intelligence & Analytics, Database Management Systems, Data Integration, Marketing Analytics

Sreyas Institute of Engineering and Technology Hyderabad, India

Bachelor of Technology in Computer Science ***Nov 2020 — Jun 2024***

Relevant Coursework: Machine Learning, Statistical Methods, Data Structures & Algorithms, Database Systems

TECHNICAL SKILLS

Languages & Databases: SQL (CTEs, Window Functions, Subqueries), Python (Pandas, NumPy, Scikit-learn), PostgreSQL, BigQuery, GCP

BI & Visualization: Power BI (DAX, Power Query), Tableau, Looker Studio, Excel (Pivot Tables, Power Query, VLOOKUP)

Analytics: Exploratory Data Analysis, Data Wrangling, ETL Pipelines, Data Modeling, Predictive Modeling, Customer Segmentation, Cohort Analysis, Funnel Analysis, Regression Analysis, KPI Development, Root Cause Analysis

PROFESSIONAL EXPERIENCE

Data Analyst Intern **Aug 2023 — Oct 2023**

Examinations Department, Sreyas Institute of Engineering and Technology *Hyderabad, India*

- Reduced cross-departmental data inconsistencies from ~**20% to under 5%** by consolidating 4 Excel-based systems into a standardized SQL dataset with enforced primary key validation and data quality checks across 2,500+ records.
- Saved **12–15 hrs/month** for academic advisors by replacing manual KPI consolidation with structured SQL extraction and pre-configured Excel dashboards, cutting weekly reporting time from 3–4 hrs to under **1 hr** per reporting cycle.
- Flagged 3 academic risk indicators — CGPA below 2.5, attendance under 60%, course backlogs — via cohort analysis across 500+ students, surfacing early signals of delayed progression.

PROJECTS

Payment Fraud Detection – SQL | Python | Power BI

- Detected **92%** of fraudulent transactions and reduced false positives by 40% by designing a BigQuery ETL pipeline across 50K+ transactions and applying Z-score–based velocity modeling with calibrated confidence thresholds.
- Surfaced high-risk behavioral deviations across 50K+ transactions by applying **CTE-driven** transformations and LAG/LEAD window functions to identify transaction-gap anomalies across the full dataset.
- Cut manual fraud pattern analysis from 3–4 hours to **under 5 minutes** by developing a DAX-automated Power BI dashboard for real-time anomaly monitoring and drill-through investigation.

Pricing Optimization & Market Segmentation – SQL | Python | Excel

- Revealed **48.8% revenue concentration** within the $1,000–$1,200 pricing tier by clustering 30K+ transactions across 200+ SKUs using K-Means.
- Quantified a $22.69 per-GB RAM price premium ($R^2 = 0.425$) through linear regression modeling, exposing underpriced SKUs and structural pricing inefficiencies across product tiers.
- Detected **25% margin variance** across APAC and North America through regional pricing-performance analysis, identifying geographies where realized pricing lagged demand indicators.

Clinical Workflow Analytics – SQL | Python

- Revealed **79% patient attrition** prior to diagnosis through multi-stage funnel modeling of a 5-stage care pathway built from 4K+ EHR records.
- Isolated referral-stage transitions as the primary delay driver by engineering **5** interval-based metrics in Python to quantify stage-to-stage diagnostic lag across patient pathways.
- Structured **4K+ fragmented EHR records** into a unified PostgreSQL data mart across 8 relational tables, supporting end-to-end measurement of diagnostic workflow performance.